

AYESHA SADDIQA

Phone number: (+92) 3087220863 **WhatsApp number:** (+92) 3087220863 **Home:** Lalamusa, Pakistan

Email address: 22011714-034@uog.edu.pk, ayesha.azeem125@gmail.com

LinkedIn: <http://linkedin.com/in/ayesha-saddiqa-02b35b19a>

Website: <https://ayesha-saddiqa-web.github.io/Portfolio>

ORCID: <https://orcid.org/0009-0007-4302-9873>

Google Scholar: <https://scholar.google.com/citations?hl=en&authuser=2&user=f3Ahf2MAAAAJ>

ResearchGate: https://www.researchgate.net/profile/Ayesha-Saddiqa-17?ev=hdr_xprf

RESEARCH PROFILE

MPhil graduate (distinction, ranked 2nd in cohort) and biomedical researcher specializing in the physiological and metabolic effects of dietary interventions. Brings a first-author publication record in biomarker-based phenotyping — spanning haematology, lipid metabolism, and oxidative and inflammatory status — with complementary strengths in gut-microbiome evaluation and the translational pharmacology of dietary botanicals. Grounded in biostatistics (SPSS) and actively developing Python and AI skills toward multi-omic data analysis. Motivated to advance the study of sustainable diets for human and environmental health, and to connect mechanistic biomarker research with the identification of diet–disease biomarkers and their molecular mechanism.

EDUCATION AND TRAINING

Master of Philosophy in Zoology (MPhil)

University of Gujrat, Hafiz Hayat Campus, Pakistan

[15/11/2022 - 27/12/2024]

Final Grade: CGPA: **3.86 / 4** (82%)

Thesis: Effect of Papaver Somniferum L. Supplementation on Growth Performance, Blood Profile, LFT and Tissue Redox Parameters of Male Rabbit

Master of Sciences in Zoology (MSc)

University of Gujrat, Hafiz Hayat Campus, Pakistan

[21/10/2019 - 29/07/2021]

Final Grade: CGPA: **3.59 / 4** (78.17%)

Bachelors of Science (BSc) [Zoology, Botany, Psychology]

University of Gujrat, Hafiz Hayat Campus, Pakistan

[11/07/2016 - 07/08/2018]

Final Grade: **523 / 800** (65.3%)

RESEARCH EXPERIENCE

Research Projects

1. Effects Of Pumpkin Seed Essential Oil on Reproductive Hormones and Histopathology of Testis and Ovaries of Rabbits

[Department of Zoology, University of Gujrat, Pakistan, 17/11/2025 – Continue]

Role: Investigation, Methodology, Statistical Validation. Optimized ELISA protocols for reproductive hormones and performed tissue histopathology.

2. Effect of Bottle Gourd (*Lagenaria Siceraria*) Seed Powder on Growth, Lipid Profile, Hematological, and

Tissue Redox Parameters of Rabbit

[Department of Zoology, University of Gujrat, Pakistan, 12/01/2026 – Continue]

Role: Investigation, Statistical Analysis, Supervision. Conducted growth trials, hematology and hepatic antioxidant assays.

3. Effect of Ajwain Powder (*Trachyspermum Ammi*) on Gut Microbiome, Blood Profile, and Immunity of Rabbit

[Department of Zoology, University of Gujrat, Pakistan, 04/02/2026 – Continue]

Role: Investigation, Statistical Validation. Processed gut microbiota samples and performed immune ELISAs to assess host-microbe interactions.

4. Construction of CRISPR-CAS9 Cassette for Bacterial Genome Insertion Through Homologous Recombination

[Department of Zoology, University of Gujrat, Pakistan, 02/05/2024 – 06/11/2024]

Role: Methodology. Assisted in molecular design and validation of CRISPR-Cas cassettes.

PUBLICATIONS

Peer-reviewed Publications:

1. Hassan M, **Saddiqa A**, Iqbal R, Yaseen R, Aftab K. Ethanolic chicory leaf extract enhances hematological profile, gut microbiota, and intestinal morphology in rabbits. *Open Vet. J.* 2026; 16(4): 2299-2312. <https://doi.org/10.5455/OVJ.2026.v16.i4.30>. **(2nd Author)**.
2. **Saddiqa, A.**, Yaseen, R., Hassan, M. *et al.* Bridging the Gap Between Herbs and Pharmaceuticals: The Science Behind Herbs' Medicinal Properties and Potential in Modern Drugs. *Rev. Bras. Farmacogn.* 35, 861–877 (2025). <https://doi.org/10.1007/s43450-025-00674-4>. **(1st Author)**.

Accepted Manuscripts:

3. **Saddiqa, A.**, Yaseen, R., Razia, I., Hassan, M., Aftab, K., Bibi, A. JAVAR (2026). *In vivo* Dose- and Time-Dependent Biphasic Modulation of Physiological Responses by Poppy Seed Supplementation **(1st Author)**

Manuscripts Under Review:

4. A Multifaceted Investigation of Differential and Synergistic Effects of Ginger and Garlic Essential Oils on Rabbit Growth, Liver Redox Activity, and Immunity. **(Corresponding Author)**
Yaseen, R., Hassan, M., Razia, I., **Saddiqa, A.**, Aftab, K., Bibi, A. *World Rabbit Science* (2026).

PROFESSIONAL EXPERIENCE

RESEARCH ASSISTANT

Laboratory of Animal Physiology, Hematology and Immunology, Department of Zoology, University of Gujrat [02/09/2024 – Continue]

Manage and execute in vivo experimental trials, ensuring strict adherence to laboratory protocols.

Perform comprehensive sample processing and advanced biostatistical analysis for diverse biological studies.

Contribute directly to experimental design, methodology, and manuscript preparation for peer-reviewed publications.

SUBJECT TEACHER

Dar e Arqam Schools Kharian Campus

[Sep 2018 – Sep 2019]

Delivered comprehensive lessons, designed learner-centred teaching strategies, and assessed student performance to support academic improvement.

SENIOR COORDINATOR

Dar e Arqam Schools Kharian Campus

[March 2018 – Sep 2019]

Supervised academic and administrative activities, supported faculty development, and monitored student progress to enhance academic performance.

CONFERENCE PRESENTATIONS, AWARDS AND RELEVANT TRAINING

Conference Presentations & Seminars:

Certificate of Participation in Thesis Poster Presentation Competition

[University of Gujrat, Hafiz Hayat Campus, Pakistan, 2024]

Participant at 2nd International Conference of Sciences

[Rawalpindi Women University, Pakistan, 15/11/2023]

Participant in Seminar on Molecular Genetic Landscape of Oculocutaneous Albinism

[Department of Zoology, University of Gujrat, Hafiz Hayat Campus, Pakistan, 20/12/2022]

Awards & Honors:

Honor Certificate as Scholar Council Member

[ETERNO PRESS, 14/08/2025-31/12/2025]

Silver Medalist in MPhil Zoology

[University of Gujrat, Hafiz Hayat Campus, Pakistan, 2022-2024]

2nd Position Holder in Exhibition on Insect Ecology and Bio-waste Management

[University of Gujrat, Hafiz Hayat Campus, Pakistan, 12/02/2023]

Third-place Holder in MSc Zoology

[University of Gujrat, Hafiz Hayat Campus, Pakistan, 2019-2021]

Awarded with Laptop via Merit-based Laptop Scheme

[Government of Pakistan, Prime Minister Youth Laptop Scheme, 14/04/2017]

Courses & Additional Training:

Advanced Coursework in Python (Online)

[Hunarmand Punjab, Punjab University, Lahore, Currently Enrolled]

Advanced Coursework in AI (Online)

[Hunarmand Punjab, Punjab University, Lahore, Currently Enrolled]

Workshop 104 "Nutrition Across the Preterm Spectrum: Prevention, Care, and Feeding Management"

[Nestlé Nutrition Institute (NNI) & Wyeth Nutrition Science Center, Istanbul, 09/09/2026-11/09/2026]

2-Days Molecular Docking Workshop

[SKI PLUS INTERNATIONAL, 29/08/2026-30/08/2026]

2-Days Bioinformatics Workshop

[SKI PLUS INTERNATIONAL, 19/08/2026-20/08/2026]

Sanger DNA Sequencing Data Analysis Bootcamp: Learn, Analyze & Interpret

[Center for Scientific Approaches, Pakistan, 15/08/2026-30/08/2026]

Training on Bio-risk Management

[Molecular Biology Lab, University of Gujrat, Hafiz Hayat Campus, Pakistan, 15/02/2023-16/02/2023]

E-commerce and Marketing Strategy

[Online from Onion and Technology, Pakistan, 10/01/2023]

Diploma in Graphic Design

[Institute of Computer Education, Lalamusa, 28/12/2021]

Let's Break the Chain of COVID-19 Infection

[Mohammed Bin Rashid University of Medicine and Health Sciences, UAE, 19/04/2020]

Webinars:

AI assisted data analysis and freelancing opportunities for students

[Center for Scientific Approaches, Pakistan, 12/09/2026]

Research Simplified: A Step-by-Step Guide from Data Collection to Journal Acceptance

[LMIE Organisation, 06/09/2026]

Hold and Kill: An Integrated Assay Strategy for RIPTAC, an Emerging Induced-Proximity Modality

[Promega Educational Events and ICE Bioscience, 02/09/2026]

Antimicrobial Resistance (AMR)

[ScientificHubX, 29/08/2026]

Meta Analysis in Research: A Practical Guide

[Pakistan Scientific and Technological Information Centre (PASTIC) & Pakistan Science Foundation (PSF), 27/08/2026]

HIV Awareness: Prevention, Early Detection, and Ending Stigma

[ScientificHubX, 26/08/2026]

Laboratory Management & ISO/IEC 17025 Compliance

[Center for Scientific Approaches, Pakistan, 23/08/2026]

Effective Crime Scene Response for First Responders

[Center for Scientific Approaches, Pakistan, 22/08/2026]

From Biological Sample to DNA Sequence: A Complete Laboratory Workflow

[Center for Scientific Approaches, Pakistan, 15/08/2026]

MCE Webinar on Making Cell Therapy Affordable and Accessible: Off-the-Shelf CAR γ δT Cells Generated Using CRISPR/AAV

[Virtual Mode, The Ohio State University, 22/10/2025]

International Webinar to Combat Human Malnutrition

[Hybrid Mode, Organized by University of Layyah Pakistan, HARRAN ÜNİVERSİTESİ Turkey, 12/01/2024]

Certificate of Appreciation in Webinar on Water Conservation

[Active Citizenship Project by British Council, 12/04/2021]

TECHNICAL AND LABORATORY SKILLS

***In vivo* Expertise:** Rabbit and murine model handling; surgical procedures, dose administration, ethical animal care, longitudinal monitoring, and growth and metabolic biometric assessment.

Proficient / Independent Wet-Lab Techniques: PCR; UV-Vis spectroscopy; histology, microscopy, microtomy, and quantitative histomorphometry; ultracentrifugation; homogenization; plant extraction; ELISA; hematological and biochemical profiling; microbial isolation, culture, identification, and Gram staining; antioxidant assays; gel electrophoresis; enzyme kinetics and Biosafety Level-2 risk mitigation.

Biostatistics & Digital Skills: SPSS, Statistix, Python, EndNote, Mendeley, Microsoft Office, ArcGIS, Adobe Illustrator, Inkscape, and Canva.

Academic Exposure: HPLC; FTIR; GC and GC-MS; LC-MS/MS; chromatography; DNA and RNA techniques, qPCR, and RNA extraction; database retrieval; mammalian cell culture; and Western blotting.

ADDITIONAL EXPERIENCE

E-COMMERCE BUSINESS OWNER & MARKETING MANAGER

[Feb 2024 – Dec 2024]

Managed e-commerce operations and growth for health and beauty products.

Developed and executed digital marketing strategies to improve online sales and brand visibility.

Optimized product management to enhance conversion rates and customer engagement.

FREELANCER SERVICES

Online Platforms

[Jan 2022 – July 2023]

Worked as a Scientific Content Writer and Graphic Designer, developing clear scientific articles, reports, and educational materials.

Synthesized complex scientific information into accessible content for diverse audiences.

LANGUAGE SKILLS

English: Proficient (IELTS Academic **6.5**)

[LISTENING **C1**, READING **C1**, WRITING **B2**, SPOKEN PRODUCTION **B2**, SPOKEN INTERACTION **C1**]

Urdu / Punjabi: Native/Bilingual

REFEREES

1. Associate Professor (Supervisor)

Name: Dr. Razia Iqbal Jafri

Phone number: (+92) 3338422798

Email: razia.iqbal@uog.edu.pk

Department of Zoology, University of Gujrat, Pakistan

2. Associate Professor

Name: Dr. Kiran Aftab

Phone number: (+92) 3214276576

Email: dr.kiran@uog.edu.pk

Department of Zoology, University of Gujrat, Pakistan

3. Associate Professor

Name: Dr. Syed Zakir Hussain Shah

Phone number: (+92) 3060660060

Email: dr.zakirhussain@uog.edu.pk

Department of Zoology, University of Gujrat, Pakistan